

Research Methodology and Intellectual Property Rights

Course Code : 21HS58B

MCQ Question Bank

MCQ questions

Module 1

1. Published information in a particular subject area is discussed in?
 - a) Journals
 - b) Research proposal
 - c) Literature review**
 - d) Bioethics
2. Sources of Literature review are all, except?
 - a) Books
 - b) Journals
 - c) Articles
 - d) Discussions**
3. The method that consists of collection of data through observation and experimentation, formulation and testing of hypothesis is called _____ method.
 - a) Empirical
 - b) Scientific**
 - c) Scientific information
 - d) Practical knowledge
4. All research processes start with _____
 - a) Observation**
 - b) Hypothesis
 - c) Experiments
 - d) Analysis
5. A Research is a _____
 - a) Lab experiment
 - b) Systematic and scientific inquiry**
 - c) Procedure
 - d) Hypothesis
6. Which one of the following is/are not the characteristic of research?
 - a) Systemic
 - b) Empirical
 - c) Replicable
 - d) Non-systematic**
7. The purpose of research is to:
 - a) Review or synthesize existing knowledge

- b) Provide solutions to problems
 - c) Explain new phenomenon
 - d) All of these**
8. Applied research is a type of research on the basis of _____
- a) Application**
 - b) Objective
 - c) Types of information sought
 - d) Application and objective
9. Which of the following is/are the type of research on objective point of view?
- a) Applied research
 - b) Explanatory research**
 - c) Qualitative research
 - d) Quantitative research
10. On the basis of information sought there are 2 types of research which are:
- a) Exploratory research & explanatory research
 - b) Descriptive research & correlation research
 - c) Qualitative research & quantitative research**
 - d) Basic research & applied research
11. Which of the following statement is not correct about a questionnaire?
- a) A technique for collecting data
 - b) Its quick and easy to collect results
 - c) An expensive method to collect data**
 - d) Set of questions for the purposes of a survey or statistical study
12. Which of the following is a first step of designing a questionnaire?
- a) Identify the goal of a questionnaire**
 - b) Identify a target demographic
 - c) Restrict the length of your questionnaire
 - d) Develop questions for a questionnaire
13. By writing a questionnaire you should avoid the following thing:
- a) Restrict the length of your questionnaire
 - b) Explain the purpose of the questionnaire
 - c) Ask more than one question at a time**
 - d) Put the most important questions at the beginning of your questionnaire
14. A systematic study directed toward greater knowledge or understanding of the fundamental aspects of phenomena is called
- a) Applied research
 - b) Basic research**
 - c) Explanatory
 - d) Quantitative
15. Preliminary data collection is a part of the _____ research.
- a) Qualitative

- b) Exploratory**
 - c) Explanatory
 - d) Descriptive research
- 16. Which of the following best describes quantitative research?
 - a) The collection of non-numerical data
 - b) An attempt to confirm the researcher's hypotheses**
 - c) Research that is exploratory
 - d) Research that attempts to generate a new theory
- 17. Which of the following is non-probability sampling technique?
 - a) Simple random sampling
 - b) Stratified random sampling
 - c) Snowball sampling**
 - d) Cluster sampling
- 18. A research scholar is guided by a trained scholar or _____.
 - a) Pharmacist
 - b) Supervisor**
 - c) Parents
 - d) Colleagues
- 19. Which one is not part of research methodology?
 - a) Collecting info;
 - b) Drawing conclusions
 - c) Follow up**
 - d) Publication
- 20. Developing the hypothesis is an objective of _____ study.
 - a) Analytical
 - b) Observational
 - c) Experimental
 - d) Descriptive**
- 21. What is the primary focus of research methodology?
 - a) Conducting experiments
 - b) Collecting data
 - c) Defining research problems**
 - d) Analyzing results
- 22. Which term refers to the systematic process of planning and conducting research?
 - a) Research problem
 - b) Research process**
 - c) Research method
 - d) Research approach
- 23. In the context of research, what is the role of defining the research problem?
 - a) Collecting data

- b) Identifying research methods
 - c) **Guiding the entire research process**
 - d) Analyzing results
24. What distinguishes research methods from research methodology?
- a) Research methods involve data collection; methodology involves problem definition.
 - b) Research methods focus on objectives; methodology focuses on approaches.
 - c) Research methods are broader; methodology is more specific.
 - d) **Research methods and methodology are synonymous.**
25. Which step in the research process involves the identification and formulation of specific research questions or issues to be investigated?
- a) Data analysis
 - b) **Problem selection**
 - c) Literature review
 - d) Defining the research problem

Module 2

1. What is the primary purpose of research design?
- A. Collecting data
 - B. Designing experiments**
 - C. Drawing conclusions
 - D. Establishing causation
2. Which of the following is a key feature of a good research design?
- A. Simplicity
 - B. Complexity
 - C. Replicability**
 - D. Ambiguity
3. Which principle of experimental design involves conducting multiple trials of an experiment?
- A. Principle of Randomization
 - B. Principle of Replication**
 - C. Principle of Local Control
 - D. Principle of Validity
4. In research design, what does the Principle of Randomization aim to achieve?
- A. Minimize confounding variables**
 - B. Maximize sample size
 - C. Increase external validity

- D. Ensure ethical conduct
5. Why is the Principle of Local Control important in experimental design?
 - A. To control for extraneous variables within the experimental setting**
 - B. To ensure random assignment
 - C. To increase generalizability
 - D. To enhance statistical power
 6. What is the primary role of the researcher in applying the Principle of Replication?
 - A. To manipulate independent variables
 - B. To conduct multiple experiments**
 - C. To control for confounding variables
 - D. To ensure random assignment
 7. Which of the following is NOT a basic principle of experimental design?
 - A. Principle of Randomization
 - B. Principle of Reliability**
 - C. Principle of Replication
 - D. Principle of Local Control
 8. Why is it important to apply the Principle of Replication in experimental research?
 - A. To increase the generalizability of the study
 - B. To ensure ethical conduct
 - C. To control for confounding variables
 - D. To enhance the internal validity of the study**
 9. Which term refers to the ability to apply research findings to settings outside the study itself?
 - A. External validity**
 - B. Internal validity
 - C. Reliability
 - D. Replicability
 10. What does a good research design aim to minimize?
 - A. Sample size
 - B. Extraneous variables**
 - C. Generalizability
 - D. Descriptive statistics
 11. In experimental design, what does the term "random assignment" mean?
 - A. Selecting participants without any criteria
 - B. Assigning participants to groups by chance**
 - C. Conducting experiments without a plan

- D. Repeating an experiment multiple times
12. Which of the following is a common challenge in applying the Principle of Local Control?
- A. Lack of statistical power
 - B. Difficulty in ensuring ethical conduct
 - C. Inability to control for extraneous variables**
 - D. Overemphasis on external validity
13. What role does the researcher play in optimizing a research design?
- A. Maximizing sample size
 - B. Minimizing statistical power
 - C. Balancing internal and external validity**
 - D. Ignoring ethical considerations
14. Why is it important to consider the features of a good research design during the planning phase?
- A. To increase complexity
 - B. To enhance ambiguity
 - C. To maximize generalizability
 - D. To improve study validity and reliability**
15. Which term refers to the process of applying research findings in a consistent and stable manner?
- A. Replication
 - B. Reliability**
 - C. Generalizability
 - D. Application
16. In experimental research, what does the term "blind design" refer to?
- A. The researcher is unaware of the study's objectives
 - B. Participants are unaware of the experimental conditions**
 - C. The study lacks clear research question
 - D. Random assignment is not utilized
17. What is the primary goal of using randomization in experimental design?
- A. Ensure a predetermined outcome
 - B. Minimize participant involvement
 - C. Control for confounding variables**
 - D. Maximize external validity
18. Which of the following is an example of applying the Principle of Local Control in an experiment?
- A. Using a control group

- B. Matching participants based on a specific characteristic**
 - C. Conducting multiple trials
 - D. Randomly assigning participants to groups
19. Why is it essential to conduct replications in scientific research?
- A. To minimize statistical power
 - B. To increase internal validity**
 - C. To decrease external validity
 - D. To ignore ethical considerations
20. What role does the research design play in addressing the need for research design?
- A. It is irrelevant to the research process
 - B. It guides the systematic planning of the study**
 - C. It minimizes the importance of ethical considerations
 - D. It maximizes the complexity of the study

Module 3

1. Researcher must select/prepare a sample design which should be.
- a. Reliable
 - b. Appropriate
 - c. Both a and b**
 - d. None of the above
2. The first step in developing any sample design is to clearly define
- a. Sampling Unit
 - b. Source List
 - c. Type of Universe**
 - d. Parameters of interest
3. The last step the researcher will decide the type of sample design to use is
- a. Budgetary constraint
 - b. Sampling Procedure**
 - c. Type of Universe
 - d. Both a and c
4. "Sample design must be viable in the context of funds available for the research study" is the _____ of a good sample design.
- a. Criteria
 - b. Characteristic**
 - c. Both a and b
 - d. Protocol

5. The probability sampling and non-probability sampling belong to which type of Sample design technique?
 - a. **Representation Basis**
 - b. Element Selection
 - c. Pattern Basis
 - d. Travelling Selection
6. _type of sampling is based on the concept of random selection, whereas _____ type of sampling is based on the concept of 'non-random' sampling respectively?
 - a. Collective, Non-probability
 - b. Non-probability, Collective
 - c. **Probability, Non-probability**
 - d. Non-probability, Probability
7. Probability sampling is also known as?
 - a. Random Sampling
 - b. Chance Sampling
 - c. **Both (a) and (b)**
 - d. Frequency Sampling
8. Non-probability sampling is also abbreviated as?
 - a. Deliberative Sampling
 - b. Purposive Sampling
 - c. Judgement Sampling
 - d. **All of the above**
9. Quota Sampling belongs to which type of Sampling?
 - a. Probability
 - b. **Non-Probability**
 - c. Collective
 - d. Both (a) and (b)
10. The statement given below refers to which type of Sampling design?
"It gives each element in the population an equal probability of getting into the sample; and all choices are independent of one another"
 - a. Random Sampling
 - b. Purposive Sampling
 - c. Simple Random Sampling
 - d. **Both (a) and (c)**
11. _means the process of assigning numbers to objects or observation, being a

function of the rules under which the numbers are assigned?

- a. Standard Deviation
- b. Measurement**
- c. Both (a) and (b)
- d. Arithmetic Mean

12. The scales of measurement is evaluated in terms of ____properties

- a. Physical
- b. Chemical
- c. Mathematical**
- d. Biological

13. Which of the following are the most widely used Measurement Scales?

- a. Nominal Scale, Ordinal Scale, Divisional scale, Additive scale
- b. Nominal Scale, Ordinal Scale, Interval Scale, Additive scale
- c. Nominal Scale, Ordinal scale, Additive scale, Interval Scale
- d. Nominal Scale, Ordinal Scale, Interval Scale, Ratio Scale**

14. The assignment of numbers of basketball players in order to identify them belongs to which measurement Scale?

- a. Nominal Scale**
- b. Ordinal Scale
- c. Interval Scale
- d. Ratio Scale

15. The _____scale places events in order, but there is no attempt to make the intervals of the scale equal in terms of some rule.

- a. Nominal
- b. Ordinal**
- c. Interval
- d. Ratio

16. _____is the appropriate measure of central tendency, while _____is the most widely used measure of dispersion respectively?

- a. Standard Deviation, Median
- b. Mean, Median
- c. Mean, Standard Deviation**
- d. Standard Deviation, Median

17. _____scales have an absolute or true zero of measurement?

- a. Nominal
- b. Ordinal

- c. Interval
 - d. Ratio**
18. Who must be aware about the sources of error in measurement?
- a. Researcher**
 - b. Student
 - c. Participant
 - d. None of the above
19. Which of the following are the sources of error in measurement?
- a. Respondent, Situation
 - b. Measurer, Instrument
 - c. Respondent, Instrument
 - d. Both (a) and (b)**
20. _measurement must meet the tests of validity, reliability and practicality.
- a. Active
 - b. Passive
 - c. Sound**
 - d. Both (a) and (b)

Module 4

1. Which one is the true example of primary data from the following options?
- A. Journal
 - B. Book
 - C. Census Report**
 - D. Newspaper
2. What is it called when the data is sourced from the place of origin?
- A. Secondary
 - B. Primary**
 - C. Secondary and primary
 - D. All of the above
3. What is it called when the data source is gathered and compiled with others?
- A. Primary Data
 - B. Quantitative data

C. Secondary data

D. None of the above

4. The method of collecting primary data is called –

A. Questionnaire and schedule method

B. Observation and Interview method

C. None of the above

D. Both A and B

5. Data that are collected or sourced from the internet is called –

A. Documentary sources of data

B. Internal secondary sources of data

C. External secondary sources of data

D. All of the above

6. The primary data is gathered through –

A. Survey

B. Experiment

C. Both options a and b

D. None of the options a and b

7. What comes right before the formulation of the hypothesis in the research?

A. Collection of data

B. Analysis of data

C. Selection of the tools of research

D. Statement of objectives

8. Which term in data collection describes the interviewer's inability to contact an interviewee?

A. Error of calculation

B. No response error

- C. Sampling bias
- D. None of the above

9. The limitations of a personal interview include-

- A. Personal biases
- B. Lack of space and time
- C. Both A and B**
- D. None of them

10. The method of collecting first-hand data includes-

- A. Mailed questionnaire.
- B. Personal interview
- C. Telephonic interview
- D. All of the above**

11. Which of the following statements are true about interpretation of data?

- A. Interpretation can't proceed without analysis.
- B. Analysis and interpretation are interdependent.
- C. The data to be interpreted should be homogeneous in nature.
- D. All the above.**

12. The last state of research process is

- A. Review of Literature
- B. Report Writing**
- C. Research Design
- D. Analysis of Data

13. Which of the following is an optional supplement of a research paper?

- A. Foot Note
- B. Appendix**
- C. Glossary
- D. Bibliography

14. A written format of a research work is known as

- A. Research paper
- B. Monograph
- C. Project Report
- D. Research Report**

15. The report submitted when there is a time lag between data collection and presentation of result is called_____

- A. Thesis
- B. Interim Report**
- C. Summary Report
- D. Article

16. Concept is of 2 types.

- A. Abstract and Coherent
- B. Concrete and Coherent
- C. Abstract and Concrete**
- D. None of the above

17. Various methods of collecting primary data include-

- A. Questionnaire method
- B. Indirect oral investigation
- C. Direct personal interview
- D. All of the above**

18. Among the following options which can provide accurate primary data?

- A. External source
- B. Internal source**
- C. Both of the above
- D. None of the above

19. The collection of primary data is more important than secondary data. This information is-

- A. Incorrect**
- B. Correct
- C. Depends on the researcher.
- D. None of the above

20. What are the key factors in choosing the right or correct method for research?

- A. Availability of funds and time
- B. Scope, nature and objectives of the research
- C. Both A and B**
- D. None of the above

Module 5

1. Intellectual property rights are established to protect what?
 - a) Land
 - b) Buildings
 - c) Creations of the mind - inventions, literary and artistic works, symbols, names, and images used in commerce (Answer)**
 - d) Vehicles
2. Which international organization promotes Intellectual Property rights?
 - a) The United Nations
 - b) The World Bank
 - c) The World Intellectual Property Organization (WIPO) (Answer)**
 - d) International Monetary Fund (IMF)
3. An example of a design that can be patented is:
 - a) A sequence of steps for baking a cake
 - b) The shape of a new model of car (Answer)**
 - c) Title of a book
 - d) An advertising slogan
4. Copyright law gives the owner:
 - a) The right to use the invention
 - b) The right to sell the invention
 - c) Exclusive rights to reproduce the work, prepare derivative works, distribute copies, and perform and display the work publicly (Answer)**
 - d) Rights for all the above

5. Which act governs patents in India?

- a) Patents Act, 1970 (Answer)**
- b) Trademarks Act, 1940
- c) Copyright Act, 1957
- d) Geographical Indications Act, 1999

6. What is the main purpose of India's Intellectual Property Law?

- a) To protect and promote innovation and creativity (Answer)**
- b) To restrict access to knowledge and ideas
- c) To provide a monopoly to inventors and creators
- d) To encourage public use of intellectual property

7. The process and methods of creating typographic designs can be protected by:

- a) Trademark
- b) Copyright (Answer)**
- c) Patent
- d) Design rights

8. What is the duration of patent protection in India?

- a) 10 years from the date of filing
- b) 20 years from the date of filing (Answer)**
- c) 5 years from the date of grant
- d) Lifetime of the inventor

9. What are the requirements for the novelty of an invention in India?

- a) The invention must be new and not known to the public before the date of filing
- b) The invention must be novel and non-obvious to the person skilled in the art (Answer)**
- c) The invention must be new and not obvious to the person skilled in the art
- d) The invention must be new and not known to the public before the date of filing

10. A person gains copyright ownership over a work by:

- a) Creating an original work (Answer)**
- b) Buying a book at a bookshop
- c) Borrowing the disk from a library
- d) Sending an email to someone

11. Which of these cannot be patented?

- a) An idea (Answer)**
- b) A method of operation
- c) A composition of matter
- d) A machine

12. What is patentable in India?

- a) Any invention, whether product or process
- b) Only novel and industrially applicable inventions (Answer)**
- c) only products of plant or animal origin
- d) Only computers and software

13. Legal protections allowing people to produce and control original works and inventions are known as?

- a) Property Rights
- b) Intellectual Property Rights (Answer)**
- c) Copyrights
- d) Patents

14. What is not protected by copyright law?

- a) The ideas and concepts explained in a novel
- b) The wording of a novel (Answer)**
- c) The cover art of a novel
- d) None of the above

15. What is the difference between a divisional patent and a continuation patent?

- a) A divisional patent is for an invention, while a continuation patent is for an improvement to an existing invention
- b) A divisional patent is for an improvement to an existing invention, while a continuation patent is for an invention (Answer)**
- c) A divisional patent is for a product that is not yet available in the market, while a continuation patent is for a product that is already available in the market
- d) A divisional patent is for a product that is already available in the market, while a continuation patent is for an improvement to an existing invention

16. How are patent applications filed in India?

- a) Online through the Indian patent office website
- b) Offline through a patent agent or attorney (Answer)**
- c) Through a legal framework in India
- d) Through a public tender process

17. What is a key feature that differentiates a trade secret from other types of intellectual property?

- a) It does not need registration
- b) It offers protection indefinitely**
- c) It provides commercial advantage

d) All of the above (Answer)

18. When does copyright protection generally begin?

- a) When the work is published
- b) When the copyright is registered
- c) When the work is fixed in a tangible medium of expression (Answer)**
- d) All of the above

19. Geographical indications are typically used to identify:

- a) An invention processes
- b) A work of literature or music
- c) A product as originating in a territory (Answer)**
- d) A business names

20. What is required for a work to be copyrighted?

- a) To be a creative work
 - b) To be an original work (Answer)**
 - c) To be a commercial work
 - d) To be a published work
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Question Bank

(5 Marks descriptive questions)

Module 1

Understand Level Questions (L2)

1. What is Research? Explain Objectives of Research.
2. What are different types of research? Explain any two in detail.
3. Explain difference between research method and research methodology
4. What are the features of good Research?
5. What are the steps in Research Process?
6. How will you select the research problem as a researcher?
7. Explain techniques involved in defining a research problem.
8. What do you mean by experimental research?
9. Differentiate between Qualitative and Quantitative research.
10. Define the term hypothesis. What are its characteristics?
11. Types of Sampling

Module 2

Apply Level Questions (L3)

11. Assume you are applying the principles of research design to solve a specific research problem, what do you understand by the term of research design?
12. Demonstrate the need for research design.
13. Implement the features of a good research design in each research scenario. Explain the Features of good research design.
14. Illustrate the Important concepts in research design for research.
15. Develop a research design for a study, ensuring that it incorporates key features of a good design.
16. Given a research scenario, apply the principles of replication to design an experiment.
17. Given a research scenario, apply the principles of randomization to design an experiment.
18. Given a research scenario, apply the principles of local control to design an experiment.

Module 3

Apply Level Questions (L3)

19. Determine the meaning of sampling design and the procedure in selecting a sample design required for a research study.
20. Examine the necessary steps involved in crafting a sampling design, focusing on the researcher's perspective.
21. Identify the characteristics of a robust sample design and explain their importance in ensuring a reliable and valid research study.
22. "Interpret the strengths and weaknesses of probability and non-probability sampling in research.
23. Determine the various methods and procedures on ensuring effective measurement in a research study?
24. Elaborate on writing the scales of measurement in terms of mathematical operations. Explain the nominal scale, ordinal scale, interval scale, ratio scale of measurement scales.
25. Highlight the possible sources of error in Measurements in detail.
26. Interpret on how the Sound measurement must meet the tests of validity, reliability, and practicality.
27. Illustrate the meaning of sampling design and explain the steps required in creating a sampling design.
28. Relate the differences and procedures involved in Probability and Non-probability Sampling in research.

Module 4

Understand Level Questions (L2)

29. Describe different methods of collecting primary data.
30. Explain different types of interview methods.
31. Mention the merits of telephone interviews in industrial surveys, particularly in developed regions.
32. Why interpretation is considered as basic component of research process.
33. Describe the techniques of Interpretation.
34. Explain different steps involved in report writing.
35. Describe the layout of the research report.
36. Explain the mechanics of writing a research report.
37. Explain Observation method in detail in the collection of primary data.
38. Explain the factors considered for selecting an appropriate method for data collection.
39. Types of Reports

Module 5

Understand Level Questions (L2)

39. Explain the need and scope areas for the effective implementation of Intellectual Properties (IP) and the associated rights in India.
40. Bring out the characteristics features of intellectual property.
41. Explain the different forms of Intellectual Property Rights.
42. Explain the importance of intellectual property.
43. Explain the role of patent offices in India and jurisdiction. (or) How many patent offices are there in India? What is their Jurisdiction.
44. What is not Patentable? (or) Bring Out the inventions not patentable as laid down in the Indian Patent Act, 1970.
45. Describe the procedure for filing of patents.
46. Enumerate the procedure for registration of copyright.
47. Explain about the Trademarks and rights arising from trademark registration.
48. What are the Intellectual Property issue areas? Explain them in detail and explain the International IPR laws.